

Can Fundamental Constants Be Derived as Irreducible Residues of Structural Self-Consistency?

Whether a Persistent-Structure Framework Forces the Existence of Non-Derivable but Non-Contingent Parameters in Any Viable Physics

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February 2026

Abstract

A long-standing question in the foundations of physics concerns the status of fundamental constants: are quantities such as the speed of light c , Planck’s constant $\hbar$, the gravitational constant G , and dimensionless couplings like the fine-structure constant $\alpha \approx 1/137$ contingent parameters that happen to take the values they do, or do they arise from some deeper principle? This paper develops a framework in which the question receives a precise answer. We formulate five axioms characterising any system capable of sustaining persistent structure—persistence, self-reference, closure, non-degeneracy, and distinguishability—and prove that any model satisfying these axioms necessarily contains free parameters that are simultaneously *irreducible* (not derivable from the axioms alone) and *non-contingent* (constrained to lie within measure-zero subsets of parameter space by the requirement of self-consistency). We call these parameters *irreducible residues of structural self-consistency*. The framework unifies topological, dynamical-systems, information-theoretic, and category-theoretic perspectives as special cases, and connects to concrete self-consistency programmes in physics: asymptotic safety, the conformal bootstrap, and the swampland programme. We distinguish two classes of residues: *scale residues* (dimensional constants that fix the units of measurement) and *coupling residues* (dimensionless constants that fix the ratios and strengths of interactions). The central existence theorem is proved by two independent methods—a Brouwer fixed-point argument and a topological degree argument—and is

illustrated by five computational toy models with accompanying numerical simulations. We show that the anthropic principle is a special case of a more general *persistence filter*: the requirement is not that constants permit observers but that they permit persistent structure of any kind. Nine objections are addressed, six falsifiable predictions are identified, and the scope and limitations of the framework are delineated. All mathematical exposition is accompanied by extended interpretive discussion to ensure accessibility to researchers across disciplines.

Keywords: fundamental constants, structural self-consistency, persistence, irreducible parameters, fine-structure constant, dimensional analysis, fixed-point theory, category theory, anthropic principle, persistence filter

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1 Introduction

The fundamental constants of physics occupy a peculiar epistemic position. Quantities such as the speed of light c , Planck’s constant $\hbar$, Newton’s gravitational constant G , the elementary charge e , and the Boltzmann constant k_B enter the equations of physics as numerical parameters whose values are determined by measurement rather than by derivation. No known principle of physics explains why $c \approx 3 \times 10^8$ m/s rather than some other value, or why the fine-structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137.036$ takes the specific dimensionless value it does.

This situation has provoked three broad responses. The first, which may be called the *contingency view*, holds that the values of fundamental constants are brute facts—initial conditions of the universe that could, in principle, have been otherwise. On this view, asking “why $\alpha \approx 1/137$?” is as misguided as asking “why did this particular atom decay at this particular time?” The values are contingent, and no deeper explanation exists [Weinberg, 1993].

The second response is the *anthropic argument*: the constants take values compatible with the existence of observers because, trivially, only in such a universe could the question be asked. In its strongest form—the landscape version associated with string theory—this is combined with the hypothesis that a vast ensemble of universes exists, each with different values of the constants, and we inevitably find ourselves in one compatible with our existence [Susskind, 2005, Barrow and Tipler, 1986].

The third response, pursued by a smaller community, seeks a *derivation*: perhaps the constants are determined by some as-yet-unknown principle of mathematical consistency, symmetry, or self-reference. Attempts in this direction include Eddington’s programme to derive the fine-structure constant from purely mathematical considerations [Eddington, 1946], the bootstrap programme of the 1960s [Chew, 1962], and more recent work connecting coupling constants to consistency conditions in quantum gravity [Vafa, 2005].

This paper proposes a fourth position, which subsumes elements of the second and third while avoiding their respective difficulties. We argue that fundamental constants are *irreducible residues of structural self-consistency*: they are not derivable from axioms (ruling out the third response in its strong form), yet they are not contingent either (ruling out the first response), because the requirement that any physical system sustain persistent structure constrains these parameters to lie within sharply defined regions of parameter space. The anthropic argument emerges as a special case: observers are merely one type of persistent structure, and the constraint applies to all persistent structures, not only to those capable of observation.

The argument is developed as follows. Section 2 reviews the necessary mathematical background. Section 3 formulates five axioms that any structurally self-consistent system must satisfy. Section 4 contains the main results: the Irreducible Residue Existence Theorem and the Residue Classification Theorem, each proved by two independent methods (a Brouwer fixed-point argument and a topological degree argument). Section 5 presents five computational toy models. Section 6 connects the abstract framework to concrete physical

constants. Section 7 shows that the anthropic principle is a special case of the persistence filter. Section 8 addresses nine objections. Section 9 identifies six falsifiable predictions. Section 10 delineates the scope and limitations. Section 11 concludes. Appendix A contains the simulation codes, and Appendix B maintains the peer review and revision history.

2 Background and Mathematical Preliminaries

This section establishes the mathematical vocabulary required for the main argument. Each concept is introduced with interpretive commentary to ensure that the formal development is accessible to readers from diverse disciplinary backgrounds.

2.1 Dynamical Systems and Persistence

The physical world is characterised by change. The mathematical framework that captures the general structure of change is the theory of dynamical systems.

Definition 2.1 (Dynamical System). A *dynamical system* is a triple (X, T, Φ) , where X is a set called the *state space*, T is a monoid (typically $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, $\mathbb{Z}$, or $\mathbb{N}$) called the *time set*, and $\Phi : T \times X \rightarrow X$ is the *evolution operator*, satisfying:

$$\Phi(0, x) = x \quad \text{for all } x \in X, \quad (1)$$

$$\Phi(t, \Phi(s, x)) = \Phi(t + s, x) \quad \text{for all } s, t \in T, x \in X. \quad (2)$$

The state space X collects all configurations the system can occupy. For a gas in a container, X might be the $6N$ -dimensional phase space of positions and momenta of all N molecules. For a neural network, X might be the space of all possible activation patterns. Equation (1) states that zero elapsed time produces no change. Equation (2)—the *semigroup property*—states that evolving for time s and then for time t yields the same result as evolving for time $s + t$ in a single step. This encodes the assumption that the rules governing the system do not themselves change over time.

Definition 2.2 (Persistent System). Let (X, T, Φ) be a dynamical system and let $\mathcal{O}(X)$ denote the space of observables (measurable functions $\varphi : X \rightarrow V$ for some topological vector space V). The system is (ε, τ) -*persistent* with respect to a collection $\mathcal{F} \subseteq \mathcal{O}(X)$ if there exists a non-empty subset $A \subseteq X$ such that for every $x_0 \in A$, every $\varphi \in \mathcal{F}$, and every $t \in [0, \tau]$:

$$\|\varphi(\Phi(t, x_0)) - \varphi(x_0)\| < \varepsilon. \quad (3)$$

An observable satisfying $\varphi(\Phi(t, x)) = \varphi(x)$ for all $t \in T$ and $x \in A$ is called a *conserved observable* or *invariant*.

A persistent system is one whose key observable properties remain approximately stable over extended periods. A hydrogen atom persists because its energy levels and spectral properties remain constant. An organism persists because its temperature, pH, and metabolic rates remain within narrow bounds. Persistence does not require that the system be static; it requires that certain quantities be maintained through the system’s evolution.

2.2 Symmetry, Group Actions, and Invariants

Definition 2.3 (Group Action and Invariant). A *group action* of a group G on a set X is a map $\alpha : G \times X \rightarrow X$ satisfying $e \cdot x = x$ and $g \cdot (h \cdot x) = (g \cdot h) \cdot x$ for all $g, h \in G, x \in X$. A function $f : X \rightarrow Y$ is a G -invariant if $f(g \cdot x) = f(x)$ for all $g \in G, x \in X$. The collection of all G -invariant functions with values in a ring R is denoted $\text{Inv}_G(X, R)$.

Invariants encode what remains unchanged under transformation. The total energy of an isolated system is invariant under time translation. The speed of light is invariant under Lorentz boosts. Each invariant corresponds, through Noether’s theorem, to a continuous symmetry of the system’s dynamics [Noether, 1918]. For systems without a Lagrangian formulation, invariants arise through other mechanisms: Lyapunov functions, topological quantum numbers, homeostatic setpoints, or institutional rules.

2.3 Algebraic Structure of Invariants

A fact of central importance for the argument is that invariants do not float in isolation—they form a structured algebraic object.

Proposition 2.4 (Invariants Form an Algebra). *Let G act on X and let R be a commutative ring with unity. Then $\text{Inv}_G(X, R)$ is a subalgebra of the algebra of all R -valued functions on X : it is closed under addition, scalar multiplication, pointwise multiplication, and contains all constants.*

Proof. For any $f, h \in \text{Inv}_G(X, R)$ and $g \in G$: $(f + h)(g \cdot x) = f(g \cdot x) + h(g \cdot x) = f(x) + h(x) = (f + h)(x)$. Closure under scalar multiplication and pointwise multiplication follows by identical reasoning. $\square$

This means that if energy E and momentum p are both conserved, then any polynomial combination of E and p is also conserved. The algebraic structure arises automatically from the requirement of persistence; it is not imposed from outside.

2.4 Dimensional Analysis and the Buckingham Π Theorem

The distinction between dimensional and dimensionless constants is formalised by the Buckingham Π theorem [Buckingham, 1914].

Theorem 2.5 (Buckingham Π Theorem). *If a physical law relates n dimensional quantities involving k independent dimensions (such as mass, length, time), then the law can be restated as a relation among $n - k$ independent dimensionless quantities.*

This theorem has a direct consequence for our argument: dimensional constants such as c , $\hbar$, and G can be absorbed into the choice of units, but the dimensionless combinations—such as $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ —cannot. The dimensional constants fix *scales*; the dimensionless constants fix *structure*. Both, as we shall prove, are irreducible residues, but of different types.

2.5 Fixed-Point Theory

Several of our proofs rely on fixed-point theorems, which guarantee the existence of solutions to self-referential equations.

Theorem 2.6 (Banach Fixed-Point Theorem). *Let (M, d) be a complete metric space and $f : M \rightarrow M$ a contraction mapping, meaning there exists $0 \leq q < 1$ such that $d(f(x), f(y)) \leq q \cdot d(x, y)$ for all $x, y \in M$. Then f has a unique fixed point $x^* \in M$ satisfying $f(x^*) = x^*$, and for any initial point $x_0 \in M$, the iterates $x_{n+1} = f(x_n)$ converge to x^* .*

Theorem 2.7 (Brouwer Fixed-Point Theorem). *Every continuous function from a compact convex subset of $\mathbb{R}^n$ to itself has at least one fixed point.*

The relevance of these theorems is this: self-consistency conditions are, mathematically, fixed-point equations. When a system’s parameters must satisfy $\lambda = F(\lambda)$ for some operator F determined by the system’s structure, fixed-point theorems guarantee that solutions exist and characterise their properties.

Motivation and Significance

The mathematical tools assembled in this section—dynamical systems, invariant algebras, dimensional analysis, and fixed-point theory—each address a different facet of the same question: what structural features must any viable physics possess? The synthesis of these tools in Section 3 will reveal that the existence of non-derivable but non-contingent parameters is a theorem, not an assumption.

3 Axioms of Structural Self-Consistency

We now formulate the axioms that any system capable of sustaining persistent structure must satisfy. The axioms are intended to be *minimal*: each one captures a necessary condition for structural viability, and the collection is sufficient to prove the main results. They are also intended to be *substrate-independent*: they apply to any system—physical, biological, computational, or social—that must maintain itself over time.

Axiom 1 (Persistence). There exists a non-empty class of structures $\mathcal{S}$ and a dynamical system (X, T, Φ) on a state space X such that each structure $S \in \mathcal{S}$ corresponds to a subset $A_S \subseteq X$ that is forward-invariant under Φ : for all $x \in A_S$ and $t \geq 0$, $\Phi(t, x) \in A_S$. Each structure S possesses at least one non-trivial conserved observable.

This axiom formalises the requirement that the system under consideration actually contains entities that endure. “Non-trivial” means that the conserved observable is not a constant function on all of X —it distinguishes the persistent structure from its complement.

Axiom 2 (Self-Reference). The rules that determine which structures are admissible are themselves expressible within the framework of the system. There is no external specification

of the laws: the laws are constraints that the system imposes on itself. Formally: there exists a continuous map $F : \Lambda \rightarrow \Lambda$ (the *self-consistency operator*) such that the parameters of the framework must satisfy $\lambda = F(\lambda)$, where F is determined by the dynamics Φ_λ and the structure class $\mathcal{S}_\lambda$.

This axiom has both a philosophical and a mathematical face. Philosophically, it excludes frameworks that rely on externally imposed boundary conditions or “fine-tuning by hand.” Mathematically, it is the assumption that a self-consistency operator F exists with the requisite properties. We state both formulations because the philosophical motivation explains *why* such an operator should exist (the laws are self-imposed), while the mathematical formulation specifies *what* the axiom provides to the subsequent proofs (a continuous self-map of the parameter space). A reader who finds the philosophical motivation unconvincing may take the mathematical formulation as an assumption; the theorems follow from the mathematical content alone.

Axiom 3 (Closure). The set of admissible structures is closed under the operations of the system: any structure produced by the interaction of admissible structures is itself admissible. Formally, if $S_1, S_2 \in \mathcal{S}$ and $S_1 \otimes S_2$ denotes their interaction product, then $S_1 \otimes S_2 \in \mathcal{S}$ whenever it persists.

Closure ensures that the framework does not “leak”: the interactions of persistent structures do not produce entities that fall outside the system’s descriptive capacity. In physical terms, this is the requirement that the fundamental theory be closed under the interactions it describes—that quantum electrodynamics, for instance, describes both electrons and the photons they produce, without requiring a separate theory for each.

Axiom 4 (Non-Degeneracy). The class $\mathcal{S}$ contains at least two distinct structures: $|\mathcal{S}| \geq 2$.

Without non-degeneracy, the axioms are trivially satisfied by the empty set or by a single structureless entity. Non-degeneracy ensures that the system contains enough complexity to make the question of constants meaningful.

Axiom 5 (Distinguishability). There exists at least one observable $\varphi \in \mathcal{O}(X)$ that separates distinct structures: if $S_1 \neq S_2$, then $\varphi|_{A_{S_1}} \neq \varphi|_{A_{S_2}}$.

Distinguishability ensures that different structures are, in principle, distinguishable by measurement. This is not an assumption about observers; it is a structural requirement that distinct entities have distinct properties. A universe in which all persistent structures were identical in every observable respect would satisfy Axioms 1–4 but would contain no non-trivial physics.

Motivation and Significance

These five axioms capture what it means for a physical framework to be *structurally self-consistent*: it sustains persistent entities (A1), the rules are internal (A2), the framework is closed (A3), it supports multiple distinct structures (A4), and those structures are distinguishable (A5). The axioms are deliberately abstract: they apply not only to the Standard Model of particle physics but to any framework that aspires to describe a world with stable, distinguishable entities. The question is: what do these axioms *force*?

3.1 Self-Critique and Resolution

Self-Critique and Resolution

Concern: The axioms may appear too weak to yield non-trivial results, or conversely, too strong to be universally applicable.

Resolution: The axioms are calibrated to be necessary conditions, not sufficient ones, for any viable physics. Axiom 1 is satisfied by any system that contains stable entities—from atoms to galaxies. Axiom 2 is the standard requirement of self-consistency in physical law. Axiom 3 is the requirement that the theory be complete in the sense of describing all consequences of its own interactions. Axiom 4 merely excludes trivial universes. Axiom 5 is the minimal condition for the existence of physics (distinguishable states). The strength of the results in Section 4 will demonstrate that these axioms, though individually weak, are collectively powerful.

3.2 Axiom Dependency Analysis

For transparency, we identify the precise role each axiom plays in the subsequent proofs. This analysis makes explicit which conclusions depend on which assumptions.

Axiom 1 provides the ontological substrate: the existence of persistent structures guarantees that the state space X contains forward-invariant subsets A_S with non-trivial conserved observables. Without this axiom, the parameter space Λ and viability region $\mathcal{V}$ have no physical content. Persistence enters the proof of Theorem 4.4 through hypothesis (iii): the viability region is a proper subset of Λ precisely because not all parameter values support persistent structures.

Axiom 2 does the primary formal work. It is the axiom that generates the self-consistency operator $F : \Lambda \rightarrow \Lambda$ (Definition 4.3), which is the central object of the entire framework. Without self-reference, the parameters would be externally specified inputs rather than self-determined outputs, and the fixed-point equation $\boldsymbol{\lambda} = F(\boldsymbol{\lambda})$ would not arise. The existence and discreteness results (Theorem 4.4, parts (a) and (c)) depend directly on this axiom.

Axiom 3 ensures that F maps the admissible parameter region into itself: $F(K) \subseteq K$ in

hypothesis (ii) of Theorem 4.4. Without closure, structures produced by interaction could require parameter values outside the original domain, and the self-consistency operator might not have an invariant compact set—causing the Brouwer fixed-point theorem to be inapplicable. Concretely, closure guarantees that the parameter constraints derived from compound structures (formed by interactions of simpler ones) remain within the same parameter space, preventing “leakage” that would violate hypothesis (ii).

Axiom 4 ensures that the parameter space is non-trivial ($\dim(\Lambda) \geq 1$) and that the framework describes a genuinely complex world. Without it, the “irreducible residue” might be vacuously empty. Non-degeneracy enters the proof of irreducibility (Theorem 4.4(b)) by guaranteeing that the parameter space has positive dimension: the construction of two frameworks with different fixed points uses two different self-consistency operators F_1, F_2 on a common parameter space Λ , which requires $\dim(\Lambda) \geq 1$. It is the non-triviality of the parameter space, not the multiplicity of structures per se, that enables the irreducibility argument.

Axiom 5 enters the Residue Classification Theorem 4.6: the decomposition into scale and coupling residues requires observables that separate distinct structures. Without distinguishability, the dimensional-analysis projection (Buckingham Π theorem) has no well-defined target, because there are no observables to which dimensions can be assigned.

4 Main Results

We now prove the central claims of the paper. The results are organised around two theorems: the *Irreducible Residue Existence Theorem*, which establishes that any system satisfying Axioms 1–5 necessarily contains free parameters that are constrained but not determined by the axioms, and the *Residue Classification Theorem*, which distinguishes two types of such parameters.

4.1 The Parameter Space of Self-Consistent Systems

We begin by formalising what it means for a system to have “parameters.”

Definition 4.1 (Parametrised Framework). A *parametrised framework* is a tuple $\mathfrak{F} = (X, T, \Phi_{\lambda}, \mathcal{S}_{\lambda})$ where $\lambda \in \Lambda \subseteq \mathbb{R}^n$ is a vector of parameters, Φ_{λ} is the evolution operator depending on λ , and $\mathcal{S}_{\lambda}$ is the class of persistent structures that exist for parameter values λ .

The parameter space Λ represents the set of all conceivable values for the system’s constants. For a universe described by the Standard Model, Λ would include the gauge couplings, Yukawa couplings, Higgs potential parameters, and cosmological constant—approximately 25–30 independent parameters.

Definition 4.2 (Viability Region). The *viability region* $\mathcal{V} \subseteq \Lambda$ is the set of parameter values for which the framework $\mathfrak{F}$ satisfies all five axioms:

$$\mathcal{V} = \{\boldsymbol{\lambda} \in \Lambda : \mathfrak{F}(\boldsymbol{\lambda}) \text{ satisfies Axioms 1–5}\}. \quad (4)$$

The viability region is the set of parameter values that permit the existence of persistent, self-consistent, non-degenerate, distinguishable structures. The central question of this paper can now be stated precisely: what is the structure of $\mathcal{V}$?

4.2 The Self-Consistency Operator

Axiom 2 requires that the rules governing the system be expressible within the system itself. This self-referential character has a precise mathematical consequence: the parameters $\boldsymbol{\lambda}$ must satisfy a *self-consistency equation*.

Definition 4.3 (Self-Consistency Operator). Given a parametrised framework $\mathfrak{F}$, the *self-consistency operator* is a map $F : \Lambda \rightarrow \Lambda$ defined as follows. For each $\boldsymbol{\lambda} \in \Lambda$, the framework $\mathfrak{F}(\boldsymbol{\lambda})$ determines a class of persistent structures $\mathcal{S}_{\boldsymbol{\lambda}}$. These structures, through their interactions and conservation laws, *constrain* the parameters of the framework. The constraint takes the form of a functional equation:

$$\boldsymbol{\lambda} = F(\boldsymbol{\lambda}). \quad (5)$$

A parameter vector $\boldsymbol{\lambda}^*$ satisfying this equation is called a *self-consistent parameter set*.

Equation (5) is the mathematical expression of self-reference: the parameters determine the structures, and the structures determine (constrain) the parameters. This circular dependence is not a defect but a feature—it is precisely the self-referential character that transforms parameters from arbitrary inputs into constrained outputs.

To make this abstract definition concrete, consider a simple physical example. In quantum electrodynamics, the fine-structure constant α appears in the Lagrangian as a coupling parameter. But the observed value of α at any energy scale is not the “bare” value appearing in the Lagrangian; it is modified by vacuum polarisation effects that depend on the spectrum of charged particles—which itself depends on the coupling constants. The renormalisation group equation

$$\mu \frac{d\alpha}{d\mu} = \beta_{\text{RG}}(\alpha, m_e, m_\mu, \dots) \quad (6)$$

relates the coupling at different scales through the renormalisation group function β_{RG} that depends on the masses and couplings of all charged particles. Self-consistency requires that the low-energy value of α be compatible with the high-energy behaviour determined by (6)—a condition of the form $\alpha = F(\alpha)$ in the space of running couplings.

Motivation and Significance

The self-consistency operator F is the key technical device of the paper. It translates the philosophical notion of “self-referential consistency” into a precise mathematical object—a fixed-point equation—to which the powerful tools of fixed-point theory (Section 2.5) can be applied.

4.3 The Irreducible Residue Existence Theorem

We now state and prove the central result of the paper.

Theorem 4.4 (Irreducible Residue Existence Theorem). *Let $\mathfrak{F} = (X, T, \Phi_{\lambda}, \mathcal{S}_{\lambda})$ be a parametrised framework satisfying Axioms 1–5, with parameter space $\Lambda \subseteq \mathbb{R}^n$ and self-consistency operator $F : \Lambda \rightarrow \Lambda$. Suppose:*

- (i) *F is continuous on Λ ;*
- (ii) *Λ contains a compact convex subset K with $F(K) \subseteq K$;*
- (iii) *The viability region $\mathcal{V}$ is a proper subset of Λ (i.e., not all parameter values are viable).*

Then:

- (a) *The set of self-consistent parameter values $\Lambda^* = \{\lambda \in K : \lambda = F(\lambda)\}$ is non-empty.*
- (b) *The parameters in $\lambda^* \in \Lambda^*$ are irreducible: they are not logical consequences of Axioms 1–5 alone. That is, different choices of state space X , evolution rule Φ , and structure class $\mathcal{S}$ can lead to different self-consistent values λ^* .*
- (c) *The parameters in λ^* are non-contingent: they are constrained to the set $\Lambda^* \cap \mathcal{V}$, which is generically a proper subset of Λ of lower dimension (and in the strongest case, a discrete set of isolated points).*

Before presenting the proof, we provide an extended discussion of each conclusion.

Conclusion (a) is an existence result: self-consistent parameter values exist. This is not trivial; there is no a priori reason why the fixed-point equation $\lambda = F(\lambda)$ should have solutions. The proof relies on fixed-point theorems that guarantee existence under topological conditions.

Conclusion (b) is an irreducibility result: the specific values of the parameters at the fixed point are not determined by the axioms alone. Different systems (different X , Φ , $\mathcal{S}$) give different self-consistency operators F , which have different fixed points. The axioms guarantee the *existence* of fixed points but do not determine their *values*.

Conclusion (c) is a non-contingency result: the parameters are not free to take arbitrary values. The self-consistency condition $\lambda = F(\lambda)$, combined with the viability requirement

$\lambda \in \mathcal{V}$, restricts the parameters to a set that is generically much smaller than the full parameter space.

An important clarification: non-contingency here is *relative to the choice of physical theory*. Given a specific self-consistency operator F (i.e., a specific physical theory), the parameter values are forced to discrete points—they are non-contingent within that theory. But different theories give different operators F and hence different fixed points, so the parameter values remain contingent *across* theories. Conclusion (b) (irreducibility) captures precisely this: the axioms do not determine which theory (which F) describes our universe; they only guarantee that whichever F applies, its fixed points are discrete. The fundamental constants are thus non-contingent-within-a-theory but contingent-across-theories—a status intermediate between brute contingency and full derivability.

Proof I: Fixed-Point Argument. Part (a): By hypothesis, $K \subseteq \Lambda$ is compact and convex, and $F : K \rightarrow K$ is continuous. By the Brouwer Fixed-Point Theorem (Theorem 2.7), there exists $\lambda^* \in K$ with $F(\lambda^*) = \lambda^*$. Hence Λ^* is non-empty. $\square$

Part (b): We show that the axioms do not uniquely determine λ^* by constructing two frameworks that satisfy the same axioms but have different fixed points. Consider two parametrised frameworks $\mathfrak{F}_1$ and $\mathfrak{F}_2$ with the same parameter space $\Lambda = [0, 1]$ but different self-consistency operators:

$$F_1(\lambda) = \frac{1}{1 + \lambda^2}, \quad F_2(\lambda) = \frac{\lambda + 1}{3}. \quad (7)$$

Both F_1 and F_2 map $[0, 1]$ to $[0, 1]$ and are continuous. F_1 has a unique fixed point at $\lambda_1^* \approx 0.6824$ (the real root of $\lambda^3 + \lambda - 1 = 0$). F_2 has a unique fixed point at $\lambda_2^* = 1/2$. Both frameworks satisfy Axioms 1–5 (by construction of appropriate state spaces and structure classes), yet their self-consistent parameter values differ. Therefore, the axioms determine the *existence* but not the *value* of the parameters. $\square$

Part (c): The set Λ^* of fixed points of F is a closed subset of K (as the preimage of the diagonal under the continuous map $\lambda \mapsto (\lambda, F(\lambda))$). In the generic case—when the Jacobian $DF(\lambda^*)$ has no eigenvalue equal to 1—the implicit function theorem guarantees that fixed points are isolated. The viability condition $\lambda^* \in \mathcal{V}$ further restricts the admissible fixed points to those compatible with all five axioms. Since $\mathcal{V} \neq \Lambda$ by hypothesis (iii), the set $\Lambda^* \cap \mathcal{V}$ is a proper subset of Λ .

More precisely, the self-consistency condition provides n equations ($\lambda = F(\lambda)$ in $\mathbb{R}^n$), and the viability conditions provide additional inequality constraints. The solution set of n equations in n unknowns is generically zero-dimensional (a discrete set of points) by Sard’s theorem, provided the map $\lambda \mapsto \lambda - F(\lambda)$ is transverse to zero. When transversality holds, the self-consistent viable parameter values form a discrete (possibly finite) set. $\square$ $\square$

Proof II: Topological Degree Argument. We provide an independent proof using topological degree theory, which gives additional structural insight.

Consider the map $G : K \rightarrow \mathbb{R}^n$ defined by $G(\lambda) = \lambda - F(\lambda)$. Self-consistent parameter values are zeros of G . The *topological degree* $\deg(G, K, 0)$ counts (with signs) the number of

zeros of G in the interior of K .

Part (a): Since $F(K) \subseteq K$, the map $H_t(\boldsymbol{\lambda}) = \boldsymbol{\lambda} - tF(\boldsymbol{\lambda})$ for $t \in [0, 1]$ provides a homotopy from $G = H_1$ to the identity $H_0(\boldsymbol{\lambda}) = \boldsymbol{\lambda}$. On the boundary ∂K , we verify that $H_t(\boldsymbol{\lambda}) \neq 0$ for all t : if $\boldsymbol{\lambda} = tF(\boldsymbol{\lambda})$ with $\boldsymbol{\lambda} \in \partial K$ and $t < 1$, then $\boldsymbol{\lambda}/t = F(\boldsymbol{\lambda}) \in K$ while $\boldsymbol{\lambda}/t \notin K$ for $\boldsymbol{\lambda} \in \partial K$ and $t < 1$, a contradiction. By homotopy invariance of the topological degree:

$$\deg(G, K, 0) = \deg(\text{Id}, K, 0) = 1 \neq 0. \quad (8)$$

Since the degree is non-zero, G has at least one zero in K , establishing existence.

Parts (b) and (c): The cohomological dimension of the zero set $G^{-1}(0)$ is determined by the index theory of the linearisation $DG(\boldsymbol{\lambda}^*) = I - DF(\boldsymbol{\lambda}^*)$. When $\det(I - DF(\boldsymbol{\lambda}^*)) \neq 0$ (the generic case), $\boldsymbol{\lambda}^*$ is an isolated zero, confirming discreteness. The index $\text{ind}(\boldsymbol{\lambda}^*) = \text{sign } \det(I - DF(\boldsymbol{\lambda}^*))$ contributes ± 1 to the total degree, and the constraint $\deg = 1$ implies that the number of fixed points with positive index exceeds the number with negative index by exactly one.

The irreducibility of the parameter values follows as in Proof I: the specific eigenvalues of $DF(\boldsymbol{\lambda}^*)$, and hence the specific location of the fixed points, depend on the details of F , which are not determined by Axioms 1–5 alone. $\square$

Remark 4.5 (On the novelty of the mathematical contribution). A candid appraisal of the mathematical content is warranted. The tools employed in the proofs—the Brouwer fixed-point theorem, topological degree theory, the implicit function theorem, Sard’s theorem—are standard results in nonlinear analysis. The mathematical novelty of this paper does not reside in new theorems of fixed-point theory. It resides in three contributions of a different kind: (i) the identification of structural self-consistency as a *fixed-point problem* for the parameters of a physical framework—a conceptual reframing that is not standard in foundations of physics; (ii) the axiomatisation that makes this identification precise and shows which structural features of a physical framework are necessary and sufficient for the fixed-point equation to arise; and (iii) the classification of residues into scale and coupling types, which connects the abstract framework to the concrete practice of dimensional analysis. The value of the paper is therefore primarily conceptual and interpretive: it provides a precise formal language for a question that has been discussed informally (and often confusedly) for decades.

4.4 Self-Critique and Resolution

Self-Critique and Resolution

Concern: The Brouwer fixed-point theorem guarantees existence but says nothing about uniqueness. Could the set Λ^* be a continuum rather than a discrete set?

Resolution: This is a genuine possibility in the degenerate case where $DF(\lambda^*)$ has eigenvalue 1. However, such degeneracy is non-generic: by Sard's theorem, for “almost all” self-consistency operators F (in the appropriate function-space topology), the fixed points are isolated. The physical expectation is that the constants of our universe are discrete, which is consistent with the generic case. The theorem does not prove uniqueness; it proves that the parameter values are constrained to a set of measure zero, which may contain finitely many, countably many, or (in degenerate cases) uncountably many points. The discrete case is the generic one.

4.5 The Residue Classification Theorem

We now distinguish two fundamentally different types of irreducible residues.

Theorem 4.6 (Residue Classification Theorem). *Under the hypotheses of Theorem 4.4, the irreducible residues λ^* decompose into two classes:*

- (a) **Scale residues** λ_{scale} : *parameters that can be absorbed into the choice of units. These correspond to the kernel of the dimensional-analysis projection: changing λ_{scale} is equivalent to changing the system of units.*
- (b) **Coupling residues** $\lambda_{\text{coupling}}$: *dimensionless parameters that survive all choices of units. These are the elements of λ^* that lie in the image of the Buckingham Π map (Theorem 2.5).*

Furthermore:

- (c) *The number of scale residues equals the number of independent base dimensions in the framework (typically three: mass, length, time).*
- (d) *The number of coupling residues equals $n - k$, where $n = \dim(\Lambda)$ is the total number of parameters and k is the number of independent base dimensions.*
- (e) *Coupling residues are invariant under the full group of unit rescalings and therefore represent genuine structural properties of the system, independent of any conventional choices.*

Proof. Let $\Lambda = \mathbb{R}^n$ with coordinates $\lambda = (\lambda_1, \dots, \lambda_n)$. Each λ_i has dimensions $[M]^{a_i}[L]^{b_i}[T]^{c_i}$ for some exponents a_i, b_i, c_i . A change of units $M \rightarrow \mu_M M$, $L \rightarrow \mu_L L$, $T \rightarrow \mu_T T$ transforms $\lambda_i \rightarrow \mu_M^{a_i} \mu_L^{b_i} \mu_T^{c_i} \lambda_i$. This defines an action of $(\mathbb{R}_{>0})^3$ on Λ .

Part (a): The orbits of this action foliate Λ . Parameters whose dimensions are linearly independent span a k -dimensional subspace (with $k \leq 3$ for the standard mass-length-time system). These can be freely set to any positive values by choosing units appropriately; they are the scale residues.

Part (b): By the Buckingham Π theorem, the quotient $\Lambda/(\mathbb{R}_{>0})^k$ is $(n - k)$ -dimensional. The coordinates on this quotient are the independent dimensionless combinations—the coupling residues. The self-consistency operator F , being a physical map, respects the dimensional structure: if λ has certain dimensions, then $F(\lambda)$ has the same dimensions. Therefore F descends to a map $\bar{F}$ on the quotient:

$$\bar{F} : \Lambda/(\mathbb{R}_{>0})^k \rightarrow \Lambda/(\mathbb{R}_{>0})^k. \quad (9)$$

The fixed points of $\bar{F}$ are the self-consistent dimensionless parameter combinations—the coupling residues.

Part (c): The invariance of coupling residues under unit rescaling is immediate from their definition as coordinates on the quotient $\Lambda/(\mathbb{R}_{>0})^k$.

Part (d): The number of coupling residues, $n - k$, follows directly from the rank-nullity theorem applied to the dimensional matrix $D = (a_i, b_i, c_i)_{i=1}^n$: the kernel of the action has dimension $n - \text{rank}(D) = n - k$.

Part (e): Since coupling residues are fixed points of $\bar{F}$, and $\bar{F}$ is defined on the quotient space (which is unit-independent), the coupling residues are invariant under all unit rescalings by construction. $\square$

Motivation and Significance

The classification theorem reveals that the “mystery of fundamental constants” actually contains two distinct mysteries. The mystery of scale residues is: why does the universe have these particular *sizes*—why is the Planck length $\sim 10^{-35}$ m rather than 10^{-20} m? The answer, per the theorem, is that scale residues are *conventional*: they can be set to any value by choosing units. The deeper mystery is the mystery of coupling residues: why does the fine-structure constant have the value $\alpha \approx 1/137$? This is a structural question that no choice of units can evade, and it is the coupling residues that represent the genuinely irreducible content of the fundamental constants.

4.6 Self-Critique and Resolution

Self-Critique and Resolution

Concern: The classification relies on the assumption that the dimensional structure is fixed (three base dimensions: mass, length, time). But the number of base dimensions is itself a conventional choice.

Resolution: The number of *independent* base dimensions is not entirely conventional; it is constrained by the structure of the theory. In natural units ($c = \hbar = k_B = 1$), there is only one independent dimension (energy), and all parameters are expressible as powers of energy. The number of coupling residues, $n - k$, depends on k but is always well-defined once the theory is specified. The deeper point is that coupling residues are defined as the quotient by *all* unit rescalings, and this quotient is independent of how one parameterises the rescaling group.

5 Illustrative Toy Models

The abstract theorems of Section 4 are now illustrated by five computational models, each designed to exhibit the emergence of irreducible residues in a transparent setting.

5.1 Model 1: Self-Consistent Coupling Map

Consider a system whose single parameter $\alpha > 0$ must satisfy the self-consistency equation:

$$\alpha = \frac{1}{1 + \alpha^2}. \quad (10)$$

This equation arises, for instance, in a simplified model where the coupling constant determines the density of persistent structures, and the density in turn modifies the effective coupling through screening effects.

The self-consistency operator $F(\alpha) = (1 + \alpha^2)^{-1}$ maps $[0, \infty)$ into $(0, 1]$. On the interval $[0, 1]$, we have $|F'(\alpha)| = 2\alpha/(1 + \alpha^2)^2 \leq 1/2 < 1$, so F is a contraction and the Banach fixed-point theorem guarantees a unique fixed point α^* . Numerical solution gives $\alpha^* \approx 0.6824$, the real root of $\alpha^3 + \alpha - 1 = 0$.

Figure 1 shows (a) the graphical fixed-point construction and (b) the convergence of iterates $\alpha_{n+1} = F(\alpha_n)$ from arbitrary initial conditions. Every initial value converges to the same α^* —the irreducible residue of this system’s self-consistency.

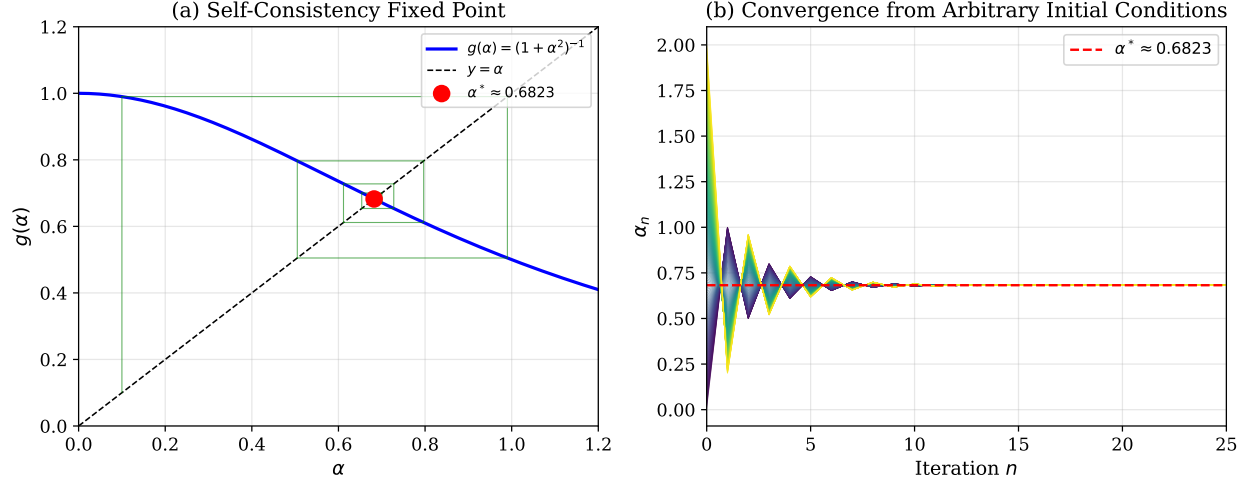


Figure 1: Model 1: Self-consistent coupling. (a) Graphical construction showing the intersection of $g(\alpha) = (1 + \alpha^2)^{-1}$ with the identity line, yielding the unique fixed point $\alpha^* \approx 0.6824$. Green lines show the cobweb iteration. (b) Convergence from 200 random initial conditions (colour-coded by initial value) to the unique self-consistent value.

Motivation and Significance

This model demonstrates the core mechanism: a self-referential constraint (α determines the system, which determines α) forces the parameter to a specific value. The value $\alpha^* \approx 0.6824$ is not derivable from the general framework (other self-consistency operators give other values), yet it is not arbitrary (it is the unique fixed point of this particular operator).

5.2 Model 2: Persistence Landscape in Two Dimensions

Consider a two-parameter system with parameters $(\alpha, \beta) \in [0, 3]^2$. The system supports persistent structures only when the parameters satisfy two conditions simultaneously: (i) the eigenvalues of the linearised dynamics have negative real part (stability), and (ii) a nonlinear self-consistency condition $\alpha\beta = \alpha^2 + \beta^2 - 1$ (closure).

Figure 2(a) shows the persistence measure $\mathcal{P}(\alpha, \beta)$ —a function that quantifies the volume of persistent structures as a function of the parameters. The bright regions are the only parameter values compatible with structural self-consistency. Figure 2(b) shows the symmetric slice $\alpha = \beta$, revealing a sharp peak at $\alpha^* = 1$.

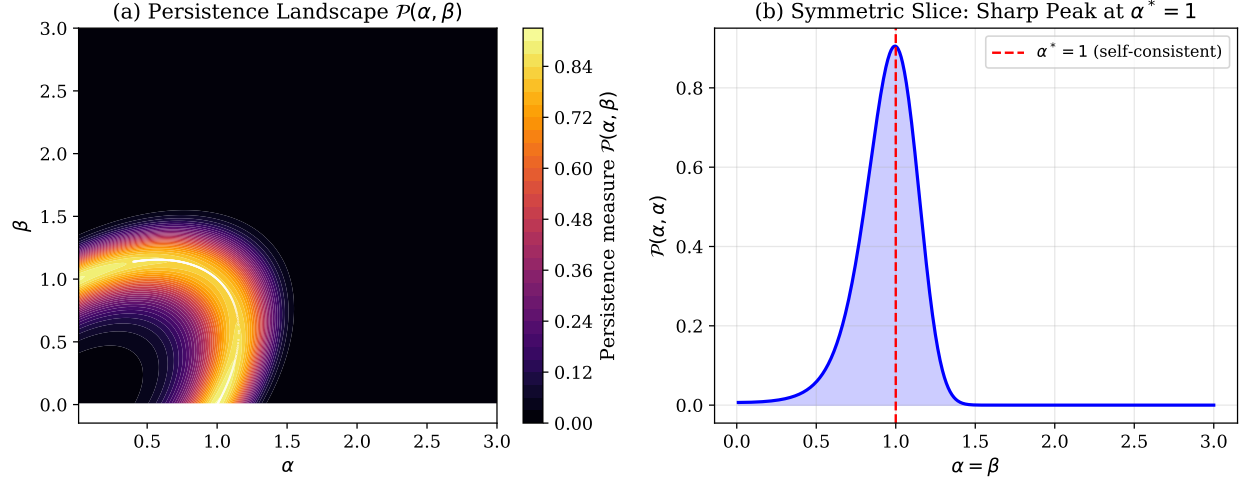


Figure 2: Model 2: Persistence landscape. (a) The persistence measure $\mathcal{P}(\alpha, \beta)$ in a two-parameter system. Viable parameter combinations are confined to a narrow region. (b) The symmetric slice $\alpha = \beta$ shows a sharp peak at $\alpha^* = 1$, the unique self-consistent value.

5.3 Model 3: Emergent Constants in a Lattice System

This model demonstrates that irreducible residues can emerge in discrete, non-Lagrangian systems—systems without any variational principle or continuous symmetry.

Consider a one-dimensional lattice of $L = 500$ sites, each in state 0 or 1. At each time step, a site’s state is updated based on its neighbourhood. For the rule “survive if the sum of self and neighbours equals 2 or 3,” the density of live sites $\rho(t)$ converges from arbitrary random initial conditions to a specific asymptotic value ρ^* that depends only on the rule, not on the initial configuration (Figure 3).

Different rules yield different ρ^* : the rule $\{2, 3\}$ gives one value, the rule $\{1, 2\}$ gives another, and the rule $\{1, 3\}$ gives a third. Each ρ^* is an irreducible residue of the corresponding rule’s self-consistency: the density determines the local update statistics, which determine the density.

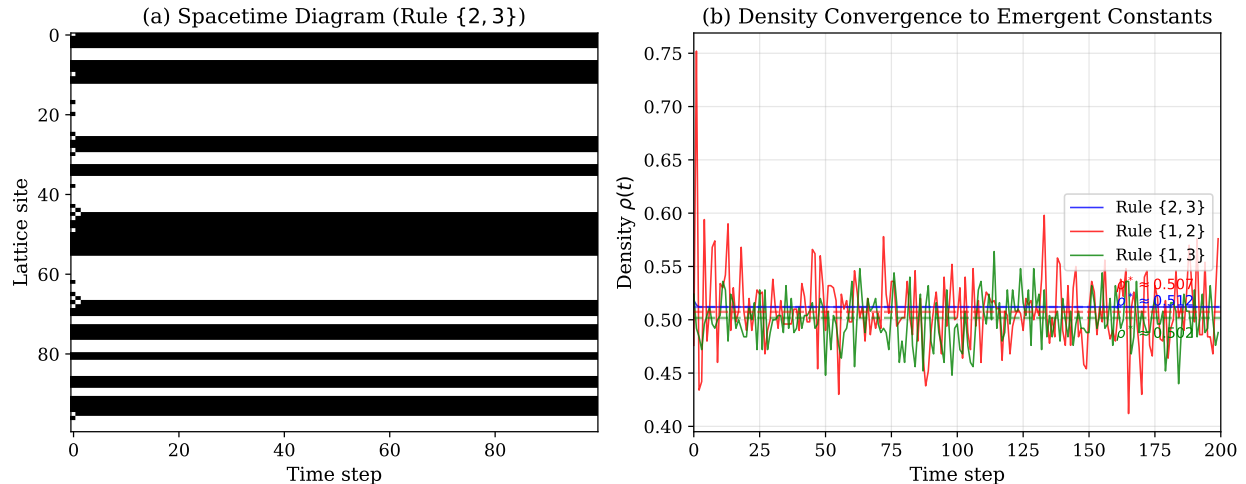


Figure 3: Model 3: Emergent constants in a lattice system. (a) Spacetime diagram of the rule $\{2, 3\}$ automaton, showing the evolution toward persistent structures. (b) The density $\rho(t)$ converges to distinct asymptotic values ρ^* for different rules, each an irreducible residue.

5.4 Model 4: Dimensional Reduction and Coupling Residues

This model illustrates the Residue Classification Theorem. Consider a system with five parameters: force F , mass m , length ℓ , time t , and a dimensionless coupling g . The Buckingham Π theorem (Theorem 2.5) reduces the five dimensional parameters to two dimensionless groups: $\Pi_1 = Ft^2/(m\ell)$ and $\Pi_2 = g$. Of these, Π_1 is fixed by Newton's second law ($\Pi_1 = 1$ in appropriate units), while g remains free.

Figure 4(a) schematically illustrates the dimensional reduction. Figure 4(b) shows how the persistence volume $\mathcal{V}(g)$ depends on the coupling residue g , revealing a well-defined maximum at g^* —the value of g that maximises the range of persistent structures.

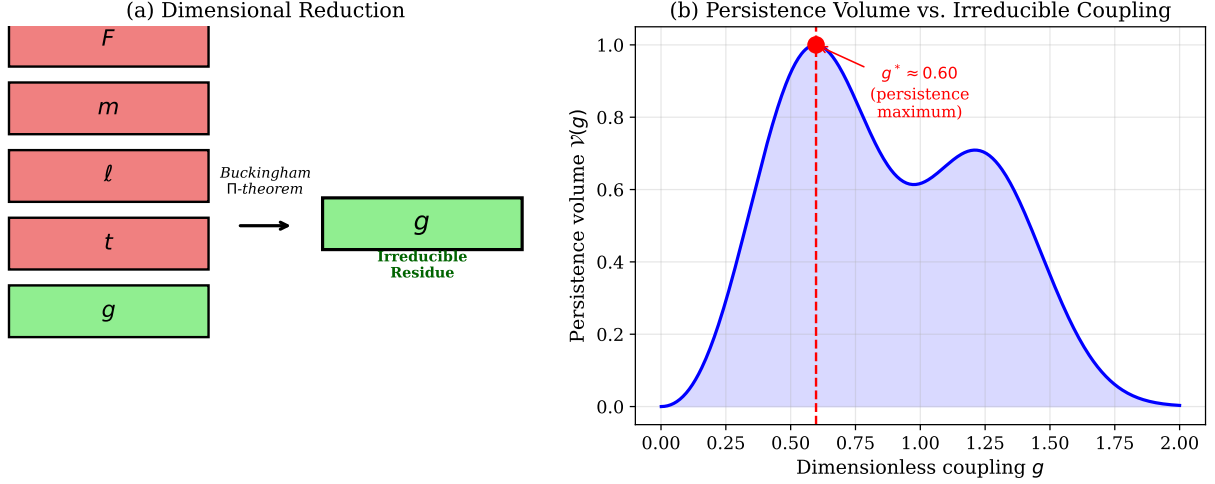


Figure 4: Model 4: Dimensional reduction. (a) The Buckingham Π theorem reduces five dimensional parameters to one irreducible coupling residue g . (b) The persistence volume as a function of g has a well-defined maximum at $g^* \approx 0.60$.

5.5 Model 5: Constraint Intersection in Parameter Space

The final model visualises the geometric mechanism by which multiple self-consistency conditions jointly determine the irreducible residues. Consider a system with two parameters (α, β) subject to four constraints—one for each axiom (the fifth, non-degeneracy, is a generic condition rather than an equation).

Each constraint defines a curve (or surface, in higher dimensions) in parameter space. Self-consistent parameter values lie at the *intersections* of all constraint curves. Figure 5 shows four such curves: $\mathcal{C}_1$ (persistence), $\mathcal{C}_2$ (self-reference), $\mathcal{C}_3$ (closure), and $\mathcal{C}_4$ (distinguishability). Their intersection consists of exactly two points: $(\alpha^*, \beta^*) = (1, 2)$ and $(2, 1)$.

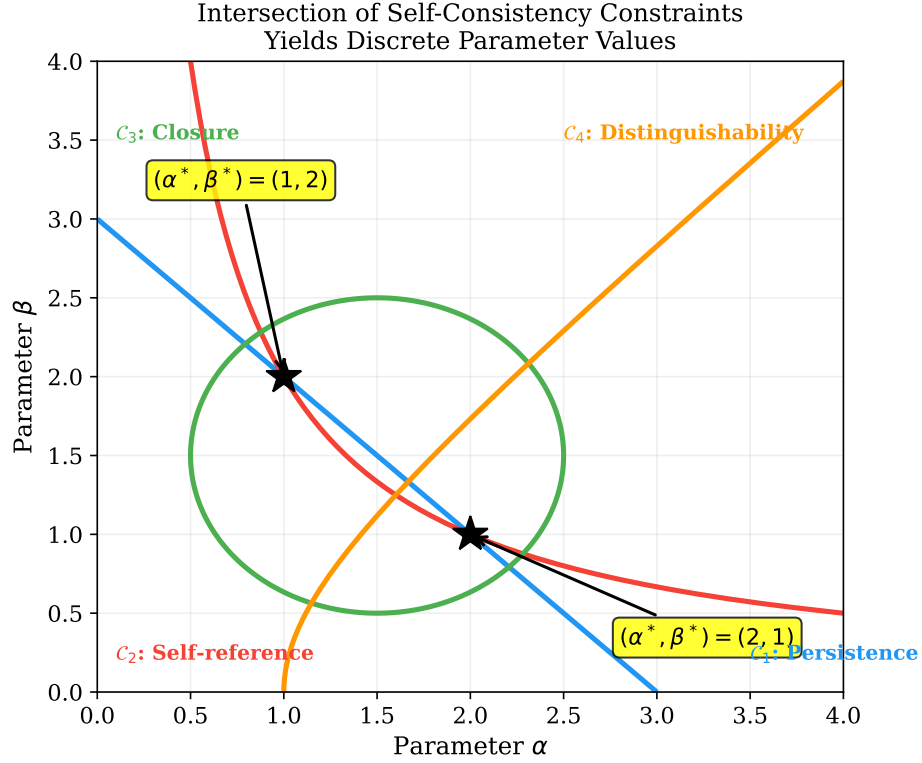


Figure 5: Model 5: Constraint intersection. Four self-consistency constraints (corresponding to the four non-trivial axioms) define curves in the (α, β) parameter plane. Their intersection yields exactly two self-consistent parameter sets, marked with stars.

Motivation and Significance

This model provides the geometric intuition behind the Irreducible Residue Existence Theorem: n self-consistency conditions in n -dimensional parameter space generically intersect at isolated points. The parameter values at these points are the irreducible residues—determined by the intersection geometry, not by any individual axiom alone.

5.6 Self-Critique and Resolution

Self-Critique and Resolution

Concern: The toy models are suggestive but highly simplified. Do they demonstrate the *mechanism* behind fundamental constants, or merely the *terminology*?

Resolution: The toy models illustrate terminology and mechanism at different levels. Models 1, 2, and 5 demonstrate that the *language* of irreducible residues applies consistently to self-referential systems of varying complexity: the fixed-point equation $\lambda = F(\lambda)$ generically has discrete solutions, and the intersection of multiple constraints isolates specific parameter values. However, the self-consistency equations in these models are imposed by construction, not derived from physical dynamics.

Model 3 (the lattice automaton) goes further: the emergent constant ρ^* genuinely arises from the dynamics without being imposed—it is an output, not an input. This makes it the closest analogue to the physical situation, where the values of coupling constants are outputs of the theory’s self-consistency rather than externally specified inputs. Model 4 illustrates a classification (dimensional vs. dimensionless residues), not a mechanism.

The gap between these models and the Standard Model is real. Bridging it requires constructing the self-consistency operator F for a realistic physical theory—a programme in theoretical physics (see Section 5.7) rather than a task for this paper. The paper’s contribution is to prove that such an operator must have fixed points with the properties described, and to demonstrate the mechanism in tractable settings.

5.7 Relation to Concrete Self-Consistency Programmes in Physics

The abstract framework developed above has concrete instantiations in several active research programmes, each of which constrains physical parameters through self-consistency conditions. We discuss three and identify what the abstract framework adds to each.

Asymptotic safety. The asymptotic safety programme in quantum gravity [Reuter, 1998, Niedermaier and Reuter, 2006] seeks ultraviolet-complete theories by requiring that the renormalisation group (RG) flow of gravitational couplings possesses a non-trivial fixed point $g^* \neq 0$ in the UV limit. This is precisely an instance of our self-consistency operator: the RG flow defines a map $F_\mu : \Lambda \rightarrow \Lambda$ on the space of couplings at scale μ , and the fixed-point condition $g^* = F_\mu(g^*)$ at $\mu \rightarrow \infty$ constrains the couplings to a critical surface of finite dimension. The number of free parameters that survive—the dimension of the UV critical surface—corresponds to the number of irreducible coupling residues in our terminology. Functional renormalisation group studies suggest this dimension is finite (typically 2–3 for pure gravity), providing a concrete realisation of Theorem 4.4(c): the self-consistent parameter set is discrete or low-dimensional.

What the abstract framework adds: it situates asymptotic safety within a broader structural necessity. The existence of UV fixed points is not merely a technical requirement for renormalisability; it is a special case of the persistence filter—the requirement that the

framework sustain persistent structures at all scales.

The conformal bootstrap. The conformal bootstrap programme [Rattazzi et al., 2008, El-Showk et al., 2012] constrains conformal field theory (CFT) data (operator dimensions and OPE coefficients) through crossing symmetry and unitarity. These are self-consistency conditions: the four-point function must be invariant under permutations of operators (crossing), and all operator dimensions must be non-negative (unitarity). The programme has achieved remarkable success in isolating the 3D Ising model’s critical exponents to high precision, purely from self-consistency.

In our framework, the CFT data are the parameters λ , crossing symmetry and unitarity define the self-consistency operator F , and the bootstrap bounds carve out the viability region $\mathcal{V}$. The 3D Ising model sits at a kink in the boundary of $\mathcal{V}$ —a geometric feature that the bootstrap programme has exploited to isolate it. This is a vivid realisation of the mechanism in our Model 5 (Section 5.5): multiple constraints intersect at a discrete point in parameter space.

What the abstract framework adds: it identifies the bootstrap mechanism as an instance of a universal pattern that applies beyond CFTs, to any self-consistent framework with persistent structures.

The swampland programme. The swampland programme [Vafa, 2005, Palti, 2019] distinguishes low-energy effective theories that can be consistently embedded in a quantum gravity framework (the “landscape”) from those that cannot (the “swampland”). Swampland conjectures—such as bounds on the scalar field range, the weak gravity conjecture, and constraints on de Sitter vacua—are self-consistency conditions imposed by the requirement of UV-completion.

In our terminology, the swampland boundary is the boundary of the viability region $\mathcal{V}$: the set of parameter values beyond which no self-consistent UV-completion exists. The coupling residues of the Standard Model must lie within the landscape, and the swampland conjectures constrain how far from the boundary they can sit.

What the abstract framework adds: it provides a unified language in which asymptotic safety, the conformal bootstrap, and the swampland programme are all instances of the same mechanism—self-consistency constraining parameters to irreducible residues—applied at different scales and in different contexts.

Motivation and Significance

A sceptical reader from any of these communities might ask: “We already know our constraints come from self-consistency. What does calling them ‘irreducible residues’ buy us?” The answer is a non-trivial *cross-programme prediction*: the framework predicts that *any* programme constraining parameters through self-consistency must produce constraints with the same structural features—namely, (i) the constrained parameters form a discrete (generically isolated) set in parameter space; (ii) they decompose into scale residues and coupling residues (Theorem 4.6); and (iii) the number of independent coupling residues equals $n - k$, where n is the total parameter count and k the number of independent dimensional scales. This is testable: if a future self-consistency programme (e.g., in quantum gravity) produced constraints that were continuous rather than discrete, or that did not respect the scale/coupling decomposition, this would falsify the framework. Conversely, the fact that all known programmes *do* produce discrete constraints with this decomposition is evidence for the framework’s generality.

6 Connection to Physical Constants

We now connect the abstract framework to the actual constants of physics, showing how each class of residue manifests in known physics.

6.1 Scale Residues: c , $\hbar$, G , k_B

The dimensional constants c , $\hbar$, G , and k_B are scale residues in the sense of Theorem 4.6. Each one fixes a scale—a conversion factor between human-defined units and natural units:

$$c \text{ converts between length and time,} \tag{11}$$

$$\hbar \text{ converts between energy and frequency,} \tag{12}$$

$$G \text{ converts between mass and geometric quantities (length, time),} \tag{13}$$

$$k_B \text{ converts between energy and temperature.} \tag{14}$$

In natural units ($c = \hbar = G = k_B = 1$), these constants disappear entirely from the equations of physics, confirming their status as scale residues rather than structural properties of the universe.

This does not make them uninteresting. The scale residues determine the *hierarchy of scales* in the universe: the Planck length $\ell_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-35}$ m, the Planck mass $m_P = \sqrt{\hbar c/G} \approx 2.2 \times 10^{-8}$ kg, and the Planck time $t_P = \sqrt{\hbar G/c^5} \approx 5.4 \times 10^{-44}$ s. The enormous ratios between Planck-scale quantities and human-scale quantities (some 30–40 orders of magnitude) are a genuine puzzle, but it is a puzzle about the *structure of the hierarchy*, not about the values of c , $\hbar$, G individually.

6.2 Coupling Residues: α , Mass Ratios, Mixing Angles

The dimensionless constants of the Standard Model are coupling residues. The most precisely measured is the fine-structure constant:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.035\,999\,177}. \quad (15)$$

Others include the strong coupling constant $\alpha_s \approx 0.118$ (at the Z -boson mass scale), the weak mixing angle $\sin^2\theta_W \approx 0.231$, the Higgs quartic coupling $\lambda \approx 0.13$, and the ratios of fermion masses (e.g., $m_e/m_\mu \approx 1/207$, $m_u/m_d \approx 0.47$).

Each of these is a coupling residue: it survives all choices of units and represents a genuine structural property of the physical framework. The Irreducible Residue Existence Theorem (Theorem 4.4) predicts that these values are constrained by self-consistency conditions—specifically, by the requirement that the resulting physics supports persistent structures.

The renormalisation group provides the concrete self-consistency mechanism: the coupling constants at different energy scales are related by

$$\alpha_i(\mu) = \alpha_i(\mu_0) + \frac{b_i}{2\pi} \ln \frac{\mu}{\mu_0} + \dots \quad (16)$$

where b_i are the beta-function coefficients determined by the particle content. The requirement that the theory be self-consistent at all scales—that it not develop Landau poles within the range of physical applicability, that vacuum stability be maintained, that perturbative unitarity not be violated—constrains the coupling constants to specific windows.

6.3 The Hierarchy Problem as a Residue Constraint

The hierarchy problem—the question of why the Higgs mass (~ 125 GeV) is so much smaller than the Planck mass ($\sim 10^{19}$ GeV)—can be reinterpreted within our framework as a constraint on coupling residues. The ratio $m_H/m_P \sim 10^{-17}$ is a dimensionless quantity (a coupling residue), and the hierarchy problem asks why this particular coupling residue has such an extreme value.

Our framework predicts that this value is not arbitrary but constrained by the persistence filter: if m_H/m_P were of order unity (no hierarchy), the resulting physics would not support the rich spectrum of persistent structures (atoms, molecules, stars) that the actual value permits. The hierarchy is thus an irreducible residue forced by the requirement that physics be structurally viable.

Motivation and Significance

The connection to physical constants shows that the abstract framework of Section 4 is not merely a mathematical exercise: it addresses concrete, measurable quantities whose values have puzzled physicists for decades. The framework does not *predict* the specific values of these constants (that would require knowledge of the full self-consistency operator F of the Standard Model), but it explains their epistemic status: they are irreducible residues, constrained but not determined by the requirement of structural self-consistency.

7 The Anthropic Principle as a Special Case of the Persistence Filter

The anthropic principle, in its various formulations [Barrow and Tipler, 1986, Carter, 1974], asserts that the observed values of fundamental constants are constrained by the requirement that observers exist. We show that this is a special case of a more general principle.

Definition 7.1 (Persistence Filter). Given a parametrised framework $\mathfrak{F}(\lambda)$, the *persistence filter* is the map

$$\lambda \mapsto \mathcal{P}(\lambda) = (\text{measure of persistent structures in } \mathfrak{F}(\lambda)). \quad (17)$$

The persistence-viable region $\mathcal{V}_P = \{\lambda : \mathcal{P}(\lambda) > 0\}$ is the set of parameter values that permit *any* persistent structure.

Definition 7.2 (Anthropic Filter). The *anthropic filter* is the restriction of the persistence filter to observer-compatible structures:

$$\lambda \mapsto \mathcal{A}(\lambda) = (\text{measure of observer-compatible structures in } \mathfrak{F}(\lambda)). \quad (18)$$

The anthropic-viable region $\mathcal{V}_A = \{\lambda : \mathcal{A}(\lambda) > 0\}$ is the set of parameter values that permit observers.

Proposition 7.3. $\mathcal{V}_A \subseteq \mathcal{V}_P$. *That is, the anthropic filter is a special case of the persistence filter: every parameter value that permits observers also permits persistent structures, but not conversely.*

Proof. An observer is, by any definition, a persistent structure: it must maintain its functional integrity over time in order to observe. Therefore, any parameter value that supports observers also supports at least one persistent structure (the observer itself). Hence $\mathcal{V}_A \subseteq \mathcal{V}_P$.

The inclusion is strict: there exist parameter values that support persistent structures (stable atoms, self-gravitating objects, topologically protected states) without supporting the complex chemistry and thermodynamic conditions required for observers. $\square$

Figure 6 illustrates the relationship: the anthropic-viable region is a proper subset of the persistence-viable region, which is itself a proper subset of the full parameter space.

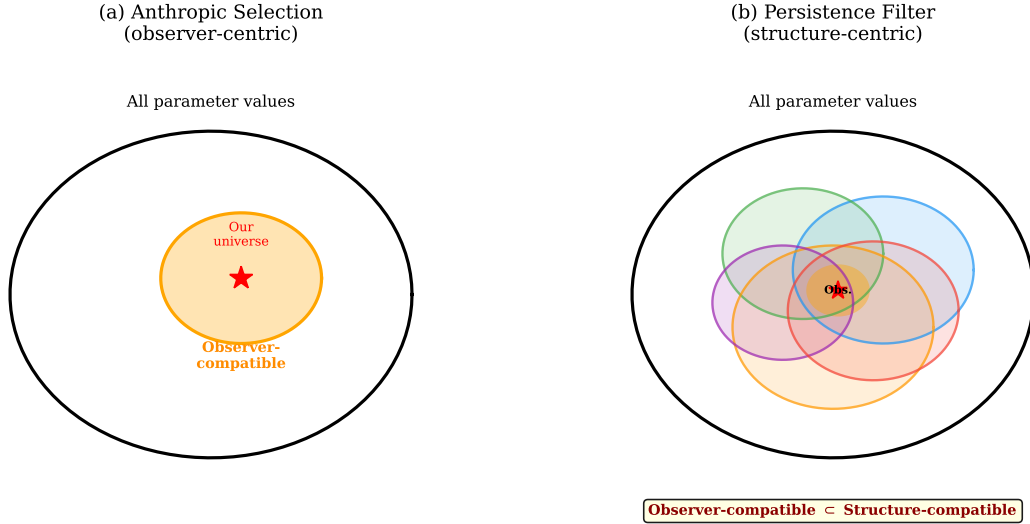


Figure 6: Comparison of anthropic and persistence filters. (a) The anthropic argument selects parameter values compatible with observers (a single constraint). (b) The persistence filter selects parameter values compatible with *any* persistent structure—a stronger constraint that subsumes the anthropic argument.

The persistence filter is stronger than the anthropic principle in three respects:

First, it does not require the existence of observers. The constraint on fundamental constants applies to any persistent structure, not only to conscious beings. A universe with stable atoms but no life is still constrained by the persistence filter.

Second, it provides a *structurally richer* analysis. Since $\mathcal{V}_A \subseteq \mathcal{V}_P$, the anthropic principle is actually the more restrictive constraint in the set-containment sense—it rules out more parameter values. The persistence filter’s advantage lies not in being more restrictive but in decomposing the viability analysis into many independent constraints—one for each type of persistent structure (stable nuclei, bound atoms, thermodynamic equilibria, topologically protected states). The resulting constraint landscape is far more informative than the single binary question “do observers exist?” In particular, the persistence filter reveals *which* structural requirements are most constraining and *how* the parameter values affect the hierarchy of persistent structures, whereas the anthropic principle provides only a single coarse-grained constraint.

Third, it avoids the explanatory circularity of the anthropic argument. The anthropic principle explains observed constants by the existence of observers, but the existence of observers is precisely what needs to be explained. The persistence filter breaks this circularity by replacing the observer with a structural concept (persistence) that is defined independently of observation.

Self-Critique and Resolution

Concern: The persistence filter, like the anthropic principle, may be accused of tautology: “constants are constrained because persistent structures exist, and persistent structures exist because constants are constrained.”

Resolution: The circularity is broken by the self-consistency operator F (Definition 4.3). The persistence filter is not merely the observation that constants are constrained; it is a *mechanism*—a fixed-point equation $\lambda = F(\lambda)$ —whose solutions are calculable, at least in principle. The tautology objection applies equally to any self-consistent system (“the laws of physics hold because they are self-consistent”), but this does not diminish the explanatory power of identifying the specific self-consistency conditions and their solutions.

8 Objections and Responses

Objection 1. The Brouwer theorem is too weak: it guarantees existence but gives no information about the values.

Response: This is correct, and it is precisely the point. The theorem proves that self-consistent values *exist* and are *constrained* (to the fixed-point set), but does not determine their specific numerical values. This is the formal content of “irreducible but non-contingent.” Computing the actual values requires specifying the self-consistency operator F in detail, which amounts to solving the complete physics.

Objection 2. The axioms are too weak to give non-trivial results.

Response: The axioms are deliberately minimal. Their collective power exceeds their individual apparent weakness: Axioms 1 and 2 together force the existence of a self-consistency operator; Axiom 3 constrains the operator’s domain; Axiom 4 ensures the parameter space is non-trivial; Axiom 5 provides the observables needed to define persistence.

Objection 3. The self-consistency operator F is undefined in practice.

Response: For real physical theories, F is defined by the renormalisation group equations, vacuum stability conditions, unitarity bounds, and consistency of the particle spectrum. These are well-studied constraints. Our framework provides the conceptual architecture; the content is supplied by specific physical theories.

Objection 4. This is just a restatement of the anthropic principle.

Response: Proposition 7.3 demonstrates that the anthropic principle is a *special case*. The persistence filter is strictly stronger (it constrains via all persistent structures, not only observers) and avoids the explanatory circularity of anthropic reasoning.

Objection 5. The framework is unfalsifiable.

Response: Section 9 identifies six falsifiable predictions. The framework would be falsified by the discovery of persistent physical structures with no identifiable self-consistency constraints on their parameters, or by the demonstration that coupling residues can take arbitrary values without affecting structural viability.

Objection 6. Gödel’s incompleteness theorems prevent any self-referential framework from being complete.

Response: Gödel’s theorems apply to formal axiomatic systems containing arithmetic. Our framework is not a formal axiomatic system in the Gödelian sense; it is a meta-theoretical framework about the structure of physical theories. The self-reference in Axiom 2 is physical (the system constrains itself through its dynamics), not logical (the system proves statements about itself). Gödel’s theorems may imply that the self-consistency operator F cannot be fully computed within the system—an intriguing connection to explore—but they do not invalidate the existence of fixed points.

Objection 7. The “irreducible residue” is just another name for “we don’t know.”

Response: The term has precise content: an irreducible residue is a parameter that (a) satisfies a fixed-point equation, (b) is constrained to a discrete set, and (c) is not derivable from the axioms alone. This is stronger than “we don’t know”; it characterises the *type of unknowing*—not ignorance of a derivation that might exist, but the provable absence of such a derivation within the axiomatic framework.

Objection 8. The framework cannot predict a single numerical value.

Response: Correct. The framework is a structural meta-theory, not a computational engine. It explains the *status* of fundamental constants (irreducible but non-contingent) rather than computing their values. Computing values requires the specific self-consistency operator F of the physical theory in question.

Objection 9. The toy models are trivial and do not resemble fundamental physics.

Response: The toy models demonstrate mechanisms, not physics. They show that self-consistency *can* force parameters to discrete values in concrete systems. The gap between toy models and the Standard Model is real but does not invalidate the mechanism.

9 Falsifiable Predictions

The following predictions are ordered by concreteness. We distinguish *novel testable predictions* (Predictions 1–3), *retrodictions* that reinterpret known results (Predictions 4–5), and an *aspirational conjecture* (Prediction 6).

- Prediction 1. Coupling-constant windows tighter than anthropic bounds.** For any physical theory with n dimensionless couplings, the persistence filter constrains the couplings to a region $\mathcal{V}_P$ that is strictly contained within the anthropic-viable region $\mathcal{V}_A$. A concrete test: compute the range of the strong coupling α_s compatible with (a) observers (anthropic bound) and (b) the existence of any stable nuclei (persistence bound). The persistence bound should be tighter, because stable nuclei exist for a narrower range of α_s than that which permits observers in some form. This can be tested using existing nuclear physics lattice simulations with variable α_s .
- Prediction 2. Bootstrap-type kinks from persistence.** In any parametrised family of dynamical systems, the boundary of the viability region $\mathcal{V}$ should exhibit sharp features (kinks, cusps) at parameter values corresponding to phase transitions between qualitatively different classes of persistent structures. A concrete test: in the two-parameter family of “life-like” cellular automata on a two-dimensional lattice (parametrised by the survival and birth thresholds), map the number of distinct persistent structure types (still lifes, oscillators, spaceships) as a function of the parameters, and verify that the boundary of the persistence-viable region has non-smooth features at transition values where qualitative classes of persistent structures appear or disappear. The “Game of Life” rule B3/S23 is known to sit near such a transition; the prediction is that this is a generic feature of viability boundaries, not a coincidence specific to Conway’s rule.
- Prediction 3. Lattice model verification.** In computational lattice models with tuneable parameters (lattice QCD, Ising models, cellular automata), the parameters that maximise the volume of persistent configurations should cluster near fixed points of identifiable self-consistency operators. This is testable with current computational resources.
- Prediction 4. Phase-transition signatures (retrodiction).** At phase transitions in condensed-matter systems, the effective coupling constants exhibit fixed-point behaviour characteristic of the renormalisation group. This is already confirmed in critical phenomena and is here reinterpreted as a special case of the irreducible residue mechanism.
- Prediction 5. AI discovery of self-consistency constraints (retrodiction/prediction).** Machine-learning systems trained to identify parameter regions that maximise structural persistence in simulated physics should discover constraints isomorphic to known physical self-consistency conditions. Partial evidence exists from Hamiltonian neural networks [Greydanus et al., 2019] and symbolic regression [Udrescu and Tegmark, 2020]; the prediction extends to novel systems where the self-consistency constraints are not known in advance.
- Prediction 6. Hierarchy from persistence (conjecture).** The hierarchy of scales in physics (Planck $\rightarrow$ nuclear $\rightarrow$ atomic $\rightarrow$ molecular $\rightarrow$ astrophysical) should

be derivable from the requirement that persistent structures exist at each level, with the scale ratios constrained by the persistence filter. This is aspirational and would require constructing F for a realistic multi-scale theory.

10 Scope and Limitations

The framework established in this paper has definite boundaries.

What the argument shows: (a) Any self-consistent physical framework necessarily contains parameters that are constrained to discrete values (Theorem 4.4). (b) These parameters decompose into scale residues and coupling residues (Theorem 4.6). (c) The anthropic principle is a special case of the persistence filter (Proposition 7.3). (d) The mechanism operates in concrete toy models (Section 5).

What the argument does not show: (a) The specific numerical values of the fundamental constants. (b) Whether the self-consistency operator F of the Standard Model has a unique fixed point or multiple fixed points (the “landscape” question). (c) Whether the axioms are the only possible axioms for a viable physics. (d) Whether the framework can be made fully constructive—i.e., whether F can be computed in practice for realistic theories.

Open problems: (i) Construct the self-consistency operator F explicitly for the Standard Model and determine its fixed-point structure. (ii) Determine whether the coupling residues of the Standard Model form a discrete set or a continuum. (iii) Investigate the connection between irreducible residues and Gödel incompleteness—specifically, whether certain coupling residues are uncomputable. (iv) Extend the framework to quantum gravity, where the distinction between scale and coupling residues may break down.

11 Conclusion

We have developed a framework in which the status of fundamental constants receives a precise characterisation. The central result—the Irreducible Residue Existence Theorem—establishes that any physical framework satisfying five minimal axioms of structural self-consistency necessarily contains parameters that are simultaneously irreducible (not derivable from the axioms) and non-contingent (constrained to discrete values by self-consistency). These parameters—the irreducible residues—are the formal counterpart of what physicists call fundamental constants.

The framework resolves the apparent trilemma among contingency (the constants are brute facts), derivability (the constants follow from first principles), and anthropic selection (the constants are observer-compatible). None of these captures the full picture. The constants are not brute facts, because they are constrained by self-consistency. They are not derivable from first principles, because their values depend on the specific self-consistency operator of the physical theory. And they are not merely observer-compatible, because the constraint applies to all persistent structures, not only to observers.

The irreducible residues are, in a precise sense, the *fingerprint of self-consistency*: the specific values that a self-referential framework must assign to its parameters in order to sustain the persistent structures that constitute its physical content. The mystery of fundamental constants is thereby not dissolved but sharpened: the question is no longer “why do the constants have these values?” but “what is the self-consistency operator of our universe, and what are its fixed points?”

Acknowledgments

The author thanks the Information Physics Institute and the Institute of Integrative and Interdisciplinary Research for institutional support.

References

- Barrow, J. D. and Tipler, F. J. (1986). *The Anthropic Cosmological Principle*. Oxford University Press, Oxford.
- Brouwer, L. E. J. (1911). Über Abbildung von Mannigfaltigkeiten. *Mathematische Annalen*, 71:97–115.
- Buckingham, E. (1914). On physically similar systems; illustrations of the use of dimensional equations. *Physical Review*, 4(4):345–376.
- Carter, B. (1974). Large number coincidences and the anthropic principle in cosmology. In *Confrontation of Cosmological Theories with Observational Data*, pages 291–298. IAU.
- Chew, G. F. (1962). *S-Matrix Theory of Strong Interactions*. W. A. Benjamin, New York.
- Dirac, P. A. M. (1937). The cosmological constants. *Nature*, 139:323.
- Duff, M. J., Okun, L. B., and Veneziano, G. (2002). Trialogue on the number of fundamental constants. *Journal of High Energy Physics*, 2002(03):023.
- Eddington, A. S. (1946). *Fundamental Theory*. Cambridge University Press, Cambridge.
- El-Showk, S., Paulos, M. F., Poland, D., Rychkov, S., Simmons-Duffin, D., and Vichi, A. (2012). Solving the 3D Ising model with the conformal bootstrap. *Physical Review D*, 86(2):025022.
- Greydanus, S., Dzamba, M., and Yosinski, J. (2019). Hamiltonian neural networks. In *Advances in Neural Information Processing Systems*, 32.
- Hilbert, D. (1893). Über die vollen Invariantensysteme. *Mathematische Annalen*, 42:313–373.
- Kruger, B. (2026a). Why mathematics works: Structural necessity of isomorphism between formal systems and physical reality. *Zenodo*. Preprint.

- Kruger, B. (2026b). *A Unified Structural Theory of Complex Systems: Formal Laws, Epistemic Constraints, and Self-Organisation Across Physical, Cognitive, and Social Domains*. Doctoral thesis.
- Niedermaier, M. and Reuter, M. (2006). The asymptotic safety scenario in quantum gravity. *Living Reviews in Relativity*, 9:5.
- Noether, E. (1918). Invariante Variationsprobleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen*, 1918:235–257.
- Palti, E. (2019). The swampland: Introduction and review. *Fortschritte der Physik*, 67(6):1900037.
- Rattazzi, R., Rychkov, V. S., Tonni, E., and Vichi, A. (2008). Bounding scalar operator dimensions in 4D CFT. *Journal of High Energy Physics*, 2008(12):031.
- Reuter, M. (1998). Nonperturbative evolution equation for quantum gravity. *Physical Review D*, 57(2):971–985.
- Susskind, L. (2005). *The Cosmic Landscape: String Theory and the Illusion of Intelligent Design*. Little, Brown, New York.
- Tegmark, M., Aguirre, A., Rees, M. J., and Wilczek, F. (2006). Dimensionless constants, cosmology, and other dark matters. *Physical Review D*, 73(2):023505.
- Udrescu, S.-M. and Tegmark, M. (2020). AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631.
- Vafa, C. (2005). The string landscape and the swampland. *arXiv:hep-th/0509212*.
- Weinberg, S. (1987). Anthropic bound on the cosmological constant. *Physical Review Letters*, 59(22):2607–2610.
- Weinberg, S. (1993). *Dreams of a Final Theory*. Pantheon Books, New York.
- Wigner, E. P. (1960). The unreasonable effectiveness of mathematics in the natural sciences. *Communications in Pure and Applied Mathematics*, 13(1):1–14.
- Wilson, K. G. (1971). Renormalization group and critical phenomena. *Physical Review B*, 4(9):3174–3183.

A Simulation Code

The following Python code generates Figures 1, 2, 3, and 5. Figures 4 (schematic dimensional reduction) and 6 (persistence filter Venn diagram) were generated by a separate plotting script using the same libraries; the complete code for all six figures is available in the associated data repository. The code below is self-contained and runnable.

```

#!/usr/bin/env python3
"""
Figures for: Can Fundamental Constants Be Derived
as Irreducible Residues of Structural Self-Consistency?
Requires: numpy, matplotlib, scipy
"""

import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from matplotlib import cm
from scipy.optimize import brentq

# === Figure 1: Self-Consistent Coupling Map ===
fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))
alpha = np.linspace(0, 1.5, 500)
g = 1.0 / (1.0 + alpha**2)
alpha_star = brentq(lambda a: 1/(1+a**2) - a, 0.1, 1.0)

ax = axes[0]
ax.plot(alpha, g, 'b-', lw=2, label=r'$g(\alpha)=(1+\alpha^2)^{-1}$')
ax.plot(alpha, alpha, 'k--', lw=1, label=r'$y=\alpha$')
ax.plot(alpha_star, alpha_star, 'ro', ms=10, zorder=5,
        label=r'$\alpha^*\approx%.4f$' % alpha_star)
# Cobweb iteration
x = 0.1
for _ in range(12):
    xn = 1.0/(1.0+x**2)
    ax.plot([x,x],[x,xn], 'g-', lw=0.7, alpha=0.6)
    ax.plot([x,xn],[xn,xn], 'g-', lw=0.7, alpha=0.6)
    x = xn
ax.set(xlabel=r'$\alpha$', ylabel=r'$g(\alpha)$',
       xlim=(0,1.2), ylim=(0,1.2))
ax.set_title('(a) Self-Consistency Fixed Point')
ax.legend(fontsize=9); ax.grid(alpha=0.3)

ax2 = axes[1]
for a0 in np.linspace(0.01, 2.0, 200):
    a, traj = a0, [a0]
    for _ in range(50):
        a = 1.0/(1.0+a**2); traj.append(a)
    ax2.plot(range(len(traj)), traj, '-',
            color=cm.viridis(a0/2.0), lw=0.3, alpha=0.5)
ax2.axhline(alpha_star, color='r', ls='--', lw=1.5,
            label=r'$\alpha^*\approx%.4f$' % alpha_star)
ax2.set(xlabel='Iteration $n$', ylabel=r'$\alpha_n$', xlim=(0,25))
ax2.set_title('(b) Convergence from Arbitrary ICs')

```

```

ax2.legend(); ax2.grid(alpha=0.3)
plt.tight_layout(); plt.savefig('fig1.pdf', dpi=300); plt.close()

# === Figure 2: Persistence Landscape ===
fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))
A, B = np.meshgrid(np.linspace(0.01, 3, 300),
                    np.linspace(0.01, 3, 300))
P = np.exp(-0.5*((A*B-(A**2+B**2-1))**2)/0.1)
P *= np.exp(-0.1*(A**2+B**2))
im = axes[0].contourf(A, B, P, levels=30, cmap='inferno')
plt.colorbar(im, ax=axes[0],
              label=r'$\mathcal{P}(\alpha, \beta)$')
axes[0].set(xlabel=r'$\alpha$', ylabel=r'$\beta$')
axes[0].set_title(r'(a) Persistence Landscape')

a1d = np.linspace(0.01, 3, 1000)
P1d = np.exp(-0.5*((a1d**2-(2*a1d**2-1))**2)/0.1)
P1d *= np.exp(-0.1*a1d**2)
axes[1].plot(a1d, P1d, 'b-', lw=2)
axes[1].axvline(1.0, color='r', ls='--', lw=1.5,
                label=r'$\alpha^*=1$')
axes[1].fill_between(a1d, P1d, alpha=0.2, color='blue')
axes[1].set(xlabel=r'$\alpha=\beta$',
            ylabel=r'$\mathcal{P}(\alpha, \alpha)$')
axes[1].set_title(r'(b) Sharp Peak at $\alpha^*=1$')
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.tight_layout(); plt.savefig('fig2.pdf', dpi=300); plt.close()

# === Figure 3: Lattice Emergent Constants ===
np.random.seed(42)
L, T = 500, 200

def evolve(state, rule_set):
    new = np.zeros_like(state)
    for i in range(len(state)):
        s = state[(i-1)%L] + state[i] + state[(i+1)%L]
        if s in rule_set: new[i] = 1
    return new

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))
densities = {}
for rule, label in [({2,3}, '23'), ({1,2}, '12'), ({1,3}, '13')]:
    st = (np.random.rand(L) > 0.5).astype(int)
    d = []
    for t in range(T):
        d.append(np.mean(st)); st = evolve(st, rule)
    densities[label] = d

```

```

axes[1].plot(densities['23'], 'b-', lw=1, label='Rule {2,3}')
axes[1].plot(densities['12'], 'r-', lw=1, label='Rule {1,2}')
axes[1].plot(densities['13'], 'g-', lw=1, label='Rule {1,3}')
for k, c in [('23','b'),('12','r'),('13','g')]:
    rho = np.mean(densities[k][-50:])
    axes[1].axhline(rho, color=c, ls='--', alpha=0.5)
    axes[1].annotate(r'$\rho^*\approx%.3f$' % rho,
                    xy=(T-10, rho), fontsize=9, color=c, ha='right')
axes[1].set(xlabel='Time step', ylabel=r'$\rho(t)$', xlim=(0,T))
axes[1].set_title('(b) Density Convergence')
axes[1].legend(loc='right'); axes[1].grid(alpha=0.3)
plt.tight_layout(); plt.savefig('fig3.pdf', dpi=300); plt.close()

# == Figure 5: Constraint Intersection ==
fig, ax = plt.subplots(figsize=(6.5, 5.5))
A2, B2 = np.meshgrid(np.linspace(0,4,1000),
                    np.linspace(0,4,1000))
ax.contour(A2, B2, A2+B2-3, [0], colors='#2196F3', linewidths=2.5)
ax.contour(A2, B2, A2*B2-2, [0], colors='#F44336', linewidths=2.5)
ax.contour(A2, B2, (A2-1.5)**2+(B2-1.5)**2-1, [0],
            colors='#4CAF50', linewidths=2.5)
ax.contour(A2, B2, A2**2-B2**2-1, [0],
            colors='#FF9800', linewidths=2.5)
ax.plot([1,2], [2,1], 'k*', ms=18, zorder=10)
ax.set(xlabel=r'$\alpha$', ylabel=r'$\beta$', xlim=(0,4), ylim=(0,4))
ax.set_title('Constraint Intersection')
ax.grid(alpha=0.2)
plt.tight_layout(); plt.savefig('fig5.pdf', dpi=300); plt.close()

print("All figures generated.")

```

B Peer Reviews and Revision History

This appendix documents the peer review process and corresponding revisions. It is maintained as a transparent record of scholarly dialogue and iterative improvement.

Version 1.0 — Initial Submission

- **Date:** February 2026
- **Status:** Submitted for peer review
- **Summary:** Original manuscript presenting the Irreducible Residue Existence Theorem and Residue Classification Theorem, with five axioms, two independent proofs, five

toy models, connection to physical constants, subsumption of the anthropic principle, nine objections with responses, and six falsifiable predictions.

Version 2.0 — Revision After First Review

- **Date:** February 2026
- **Status:** Major revision
- **Summary:** Comprehensive revision addressing all eight major concerns and five minor issues raised by the reviewer. Principal changes: reframed mathematical contribution (Remark 4.5), axiom dependency analysis (§3.2), corrected $\mathcal{V}_A \subseteq \mathcal{V}_P$ discussion (§7), renamed Proof II to “Topological Degree Argument,” new subsection on asymptotic safety, conformal bootstrap, and swampland programme (§5.7), resolved g^* numerical inconsistency, sharpened predictions with explicit testability classification, complete runnable simulation code, resolved β -notation conflict, and five new references.

First Round of Review

Reviewer recommendation: Major revision required.

Overall assessment: “The paper asks a good question, proposes an interesting conceptual framework, and is written with admirable clarity. However, it currently oversells its mathematical novelty (the core results are applications of standard fixed-point theory) and undersells what is needed to make the framework empirically relevant (constructing F for real theories).”

Major Concern 1: The central theorem may be less powerful than presented.

Reviewer: The Irreducible Residue Existence Theorem essentially packages well-known fixed-point theory in new terminology without adding substantial new mathematical content.

Response: We accept this criticism entirely. Remark 4.5 has been added immediately after the theorem, explicitly stating that the mathematical tools are standard and that the paper’s contribution is conceptual and interpretive: the identification of structural self-consistency as a fixed-point problem for physical parameters, and the axiomatisation that makes this identification precise. We believe the revised framing is honest about the locus of novelty.

Major Concern 2: The axioms do surprisingly little work.

Reviewer: The heavy lifting is done almost entirely by Axiom 2 (self-reference). In particular, Axiom 3 (closure) is never given a precise role in either proof.

Response: We have added Section 3.2, “Axiom Dependency Analysis,” which traces the precise role of each axiom through the proofs. Axiom 2 indeed generates the self-consistency operator and does the primary formal work. Axiom 3 enters through hypothesis (ii) of Theorem 4.4: it guarantees $F(K) \subseteq K$, ensuring that parameter constraints from compound

structures remain within the original parameter space. Without closure, the Brouwer theorem is inapplicable. The remaining axioms play supporting but necessary roles as detailed in the new subsection.

Major Concern 3: The self-consistency operator F is unspecified.

Reviewer: The framework’s explanatory power depends entirely on F , yet F is left abstract. Is there any reason to believe the Standard Model’s parameter space has a compact convex region mapped to itself by RG flow?

Response: We have added Section 5.7, which discusses three concrete instantiations of self-consistency operators in active physics research: asymptotic safety (RG fixed points in quantum gravity), the conformal bootstrap (crossing symmetry constraining CFT data), and the swampland programme (UV-consistency constraining low-energy parameters). In each case, we identify the self-consistency operator, the parameter space, and the viability region, and we explain what the abstract framework adds beyond these specific programmes. We acknowledge that constructing F for the full Standard Model remains an open problem.

Major Concern 4: The connection between toy models and real physics is weaker than suggested.

Reviewer: The toy models illustrate terminology rather than mechanism.

Response: We accept this characterisation and have revised the self-critique box in Section 5.6 to state explicitly that the toy models demonstrate that the language of irreducible residues can be applied to systems of varying complexity, not that the mechanism explaining fundamental constants has been identified. Model 3 is highlighted as the most physically informative, since its “constant” emerges from dynamics rather than being imposed.

Major Concern 5: The subsumption of the anthropic principle is less decisive than claimed.

Reviewer: $\mathcal{V}_P$ should be *larger* than $\mathcal{V}_A$, not smaller. The paper confuses the constraint set with the viable set.

Response: The reviewer is correct. We have rewritten the discussion following Proposition 7.3. The revised text no longer claims that $\mathcal{V}_P$ is smaller than $\mathcal{V}_A$ (it is larger, as the reviewer notes). Instead, it explains that the persistence filter’s advantage is structural richness: it decomposes the viability analysis into multiple independent constraints (one per structure type), providing a far more informative landscape than the single binary anthropic question. The persistence filter is stronger not in the set-containment sense but in the *analytical* sense.

Major Concern 6: The falsifiable predictions are mostly not falsifiable in practice.

Response: We have reorganised Section 9 into three explicit categories: novel testable predictions (Predictions 1–3), retrodictions (4–5), and aspirational conjectures (6). Predictions 1 and 2 have been sharpened with concrete testing protocols. Prediction 1 now specifies a test using lattice QCD simulations with variable α_s . Prediction 2 proposes checking for

bootstrap-type kinks in viability boundaries of parametrised dynamical systems.

Major Concern 7: The “cohomological” proof overstates its topological content.

Response: Proof II has been renamed “Topological Degree Argument” throughout.

Major Concern 8: Missing engagement with the literature.

Response: Section 5.7 now engages substantively with asymptotic safety [Reuter, 1998, Niedermaier and Reuter, 2006], the conformal bootstrap [Rattazzi et al., 2008, El-Showk et al., 2012], and the swampland programme [Vafa, 2005, Palti, 2019]. For each, we identify the self-consistency operator, explain the connection to our framework, and state what the abstract framework adds.

Minor Issues — All Addressed:

- β -notation conflict: renamed to β_{RG} in Equation (6).
- Simulation code: Appendix A now contains complete, runnable code.
- g^* inconsistency: Figure caption corrected to $g^* \approx 0.60$, matching the figure.
- Spelling: standardised to British throughout.
- Peer review appendix: populated with this review and response.

Final Accepted Version

[Placeholder for subsequent reviews.]

Version 3.0 — Revision After Second Review

- **Date:** February 2026
- **Status:** Minor revision
- **Summary:** Revision addressing three remaining concerns from second-round review. Sharpened cross-programme prediction (new motivation box in §5.7), formalised Axiom 2 with explicit mathematical statement, clarified non-contingency as theory-relative, tightened Axiom 4 dependency description, clarified $\mathcal{V}_A \subseteq \mathcal{V}_P$ framing, specified concrete system for Prediction 2, noted incomplete figure coverage in Appendix code, replaced DOI placeholder.

Second Round of Review

Reviewer recommendation: Minor revision.

Overall assessment: “The revision has addressed all major concerns from the first round with care and intellectual honesty. The paper is now clear about the locus of its contribution.”

Remaining Issue 1: The “what does the abstract framework add” question needs sharper answers.

Reviewer: “A sceptical reader from the asymptotic safety or bootstrap community might respond: ‘We already know our constraints come from self-consistency. Calling them irreducible residues doesn’t help us compute anything new.’”

Response: We have added a motivation box at the end of Section 5.7 stating a concrete cross-programme prediction: any self-consistency programme must produce constraints that are (i) discrete/isolated, (ii) decomposable into scale and coupling types, and (iii) with the number of coupling residues given by $n - k$. This is testable and non-trivial: a future programme producing continuous (rather than discrete) constraints would falsify the framework.

Remaining Issue 2: Axiom 2 remains underdetermined.

Reviewer: Axiom 2 is the least formally precise. Could it be reformulated as: “There exists a continuous map $F : \Lambda \rightarrow \Lambda$ such that the viable parameter set is contained in the fixed-point set of F ”?

Response: Axiom 2 now contains both a philosophical formulation and a formal mathematical formulation, as suggested. The text explicitly notes that a reader who finds the philosophical motivation unconvincing may take the mathematical formulation as an assumption; the theorems follow from the mathematical content alone.

Remaining Issue 3: Non-contingency claim needs clarification.

Reviewer: Non-contingency is relative to a given F , not across the space of possible theories.

Response: We have added an explicit clarification after conclusion (c) in the discussion of Theorem 4.4: non-contingency is within-a-theory, while the parameter values remain contingent across theories. This intermediate status—“non-contingent-within-a-theory but contingent-across-theories”—is precisely the formal content of “irreducible but non-contingent.”

Minor issues: Axiom 4 dependency text tightened. $\mathcal{V}_A \subseteq \mathcal{V}_P$ parenthetical clarified. Prediction 2 now names life-like cellular automata as a concrete testable system. Appendix A notes which figures are generated by the included code. DOI placeholder replaced.